

Silhouette

Project Report — Architecture, Tools, and the Path to User Testing

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Status: corpus-complete, pre-beta. The +10-per-state SIO expansion is finished — the library now holds **111 SIOs** across all six states, all committed. This report reflects verified corpus numbers and a fresh QA pass (`validate-media` , `detect-gaps` , `test-magnet-risk` , retrieval calibration).

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1 • Executive Summary

Silhouette is a personalized insight-retrieval engine for young professionals who feel stuck. A user describes their situation in a few sentences; Silhouette classifies the underlying "stuck state," retrieves one real, human-sourced insight from a curated library, explains in one line why it fits, and — when possible — links to the exact moment in a video or talk where that insight was spoken. The goal of each session is a single genuine *aha moment*, not engagement or time-in-app.

The product is built and functional: a Next.js + LangChain application with a six-state classifier, a scored retrieval engine with diversity controls, a presentation layer, a feedback loop, and a substantial back-office pipeline of roughly 45 scripts for discovering, evaluating, verifying, and promoting content. The corpus is being expanded to the threshold needed for real user testing.

111

TOTAL SIOS (97 WITH VERIFIED VIDEO)

6

STUCK STATES

~45

BACK-OFFICE TOOLS

This report does three things: (1) gives a clean overview of what Silhouette is and how it is structured; (2) inventories every tool we have built; and (3) lays out how to improve those tools and what to build next — so the project is legible again and ready to take into a closed beta.

Part 1 • What Silhouette Is

1.1 The product in one paragraph

Silhouette is context-aware insight retrieval. It is **not** a chatbot, a meditation or wellness app, a journaling tool, a content feed, or a therapy product. The user brings an emotional/situational moment; the system returns one curated, credible, human-sourced insight — a quote or passage from someone who navigated something similar — with a short, specific reason it applies. The conversation is the intake mechanism; the recommendation is the product; the curated library is the moat.

1.2 The user and the "aha moment"

The target user is a young professional (roughly 22–32) who is functional but stuck — low motivation, unclear direction, a post-achievement plateau — and who already consumes self-improvement content but can't find the right thing in the moment they need it. Success is defined narrowly and deliberately: the user enters in a stuck state, receives one insight plus a reason, and leaves with a real shift in perspective and a sense of a path forward. Usage is intentionally intermittent; the product optimizes for the quality of a single session, not retention.

1.3 The six stuck states

The corpus and classifier are organized around six soft, non-clinical "stuck states":

State	What it feels like
inaction-loop	Knows what to do, can't get themselves to do it; avoidance and resistance.
engagement-drought	Going through the motions; flat, disengaged, depleted (burnout-adjacent).
direction-collapse	Lost sense of direction or purpose; includes a "post-achievement" variant.
possibility-paralysis	Too many options; can't commit; fear of the wrong choice.
identity-transition	Identity built on a role or achievement that's gone or changing.
momentum-gap	Fell off, lost a streak, feels "behind," struggling to restart.

These are clusters for retrieval and framing, never diagnoses. Users can implicitly correct the classification.

1.4 The SIO and the resonance model

The unit of the library is a **Structured Insight Object (SIO)**: a single insight extracted from a credible human source, stored as a markdown file with a rich YAML metadata header plus a short insight body. Beyond the stuck-state tag, every SIO carries a **resonance profile** on two dimensions that drive ranking:

- **insight_type** — one of `mechanism` , `story` , `reframe` , `permission` .
- **voice_register** — e.g. direct/challenging, expert/scientific, warm/affirming, vulnerable/personal, intellectual/measured.

This lets retrieval match not just *topic* but *the kind of thing* that will land for this person in this moment.

1.5 Current status snapshot

Corpus size by state (pre-expansion baseline → target after the +10-per-state build):

State	Baseline	Target	Final (verified)
direction-collapse	10	20	20
engagement-drought	10	20	20
inaction-loop	10	20	20
identity-transition	8	18	18
momentum-gap	6	16	16
possibility-paralysis	7	17	17
Total	51	~111	111

Media coverage: **97 of 111 SIOs carry a verified video embed** (every state well above the ≥ 7 floor), **75 with confirmed timestamps**; the remaining 14 are honestly marked text/not-applicable. Verbatim-verified transcripts: **67 of 111** (the rest are honestly flagged as reconstructed). Source records: **113** (baseline 53).

Part 2 • How the Project Is Structured

2.1 Tech stack

- **App:** Next.js (App Router, React 19), TypeScript.
- **Agent:** LangChain.js (`createToolCallingAgent` + `AgentExecutor`), OpenAI `gpt-4o-mini` .
- **Embeddings / retrieval:** OpenAI `text-embedding-3-small` ; in-memory vector store with custom scored search, resonance boosts, and MMR diversity.
- **External:** Tavily (web-search fallback), YouTube Data API v3 + `yt-dlp` (media discovery & timestamp work).
- **Data:** file-based corpus (markdown SIOs + JSON source records); feedback as JSONL.

2.2 Repository layout

Path	Contents
<code>src/</code>	The running app: agent, retrieval (rag), presentation, tools, memory, feedback, API routes, UI components.
<code>corpus/</code>	The library: <code>sios/</code> (insights), <code>sources/</code> (source records), <code>candidates/</code> , <code>drafts/</code> , <code>templates/</code> , feedback log.
<code>scripts/</code>	~45 back-office tools: discovery, media/timestamp, evaluation, QA, retrieval diagnostics, feedback analysis (+ <code>lib/</code> shared code).
<code>ai/guides/</code>	~66 design docs, component specs, strategy, and phase reports.
<code>aiDocs/</code>	Canonical product docs: context, PRD, MVP, architecture.
<code>ai/reports/</code>	This report and its PDF export.

2.3 Runtime architecture (a single chat turn)

The path a message takes, end to end:

1. **Client → API.** `ChatWindow` POSTs the message + session to `/api/chat` .
2. **Feedback short-circuit.** `detectFeedbackIntent` catches short "yes" / "show me something different" replies *before* the agent runs, so feedback never retrieves against itself. Negative feedback triggers a one-time retry that excludes already-seen SIOs and prefers a different `insight_type` .
3. **State classification.** `classifyState` labels the message with a stuck state, confidence, and resonance hints; injected into the agent as a system note.

4. **Agent loop.** The LangChain agent calls `knowledge_base` → `scoredSearch` (state filter + resonance boosts + MMR diversity, excluding seen SIOs) → picks the best `insight_id` → calls `present_insight` .
5. **Presentation.** `presentInsight` validates attribution, generates the guarded "why this applies" (and bridge/credibility) lines, derives confidence/verification labels, and renders the insight markdown.
6. **Render.** `InsightCard` parses the markdown into slots (excerpt, attribution, why, source link) and renders the verified media embed when present.
7. **Feedback.** `FeedbackRow` posts a dwell signal to `/api/feedback-signal` , then the follow-up routes back through the short-circuit; all events land in `feedbackLog` .

2.4 The SIO data model

Each SIO's YAML header carries (abbreviated): identity (`insight_id` , `source_id`), source metadata (speaker, show/platform, title, date, URL), a full media block (provider, `video_id` , `embed_url` , `timestamp_start_seconds` , `display_mode` , `media_verification_status` , rights notes), classification (`primary_state_tag` , `secondary_state_tags`), resonance (`insight_type` , `voice_register` , `credibility_tier` , `intensity_level`), the content (`key_claim` , `content_summary` , attribution, the excerpt body), and review/honesty fields (`transcript_verified` , `human_review_status` , rights/usage notes). The same fields drive both retrieval ranking and honest presentation.

2.5 The four runtime agent tools

Tool	Role
<code>knowledge_base</code>	Searches the curated SIO corpus via scored, state-filtered, diversity-aware retrieval; returns ranked candidates.
<code>present_insight</code>	Turns a chosen <code>insight_id</code> into the final rendered insight and records presentation context for feedback/retry.
<code>web_search</code>	Tavily fallback when the curated library lacks a strong match (used sparingly).
<code>calculator</code>	Safe arithmetic for making any next-step quantification concrete.

Part 3 • The Tools We've Built

Beyond the running app, Silhouette has a substantial back-office: roughly 45 scripts that discover, evaluate, verify, and promote content, plus retrieval and feedback diagnostics. These are the "agents" and utilities that build and protect the corpus. They are grouped below with the npm alias where one exists.

3.1 Discovery & SIO harvesting

Tool (npm)	What it does
<code>discover-sios</code>	Orchestrator that chains the discovery stages (user-needs → gap → candidate → review → draft) and stops for human review. The "SIO harvesting agent."
<code>detect-gaps</code>	Deterministic coverage analysis by state/type/register/intensity; ranks the highest-priority gaps to fill.
<code>analyze-user-needs</code>	Maps the corpus against hypothesized user-need / query patterns to find coverage holes from the user side.
<code>find-source-candidates</code>	Turns gaps into ranked search queries across trusted source families (the "Source Scout").
<code>evaluate-candidate</code>	LLM-as-judge resonance evaluator (10 dimensions + anti-generic floor); scores candidates.
<code>score-candidate-novelty</code>	Embedding-based dedup vs. the served corpus (duplicate / similar-but-useful / novel).
<code>verify-candidate-source</code>	Deterministic honesty checks; can only <i>downgrade</i> a status, never upgrade to verified.
<code>review-candidates</code>	Builds a ranked review queue with blockers, grouped by action.
<code>draft-sio-from-candidate</code>	Writes prototype SIOs to <code>corpus/drafts/</code> only (never directly to the served corpus).
<code>promote-candidate-to-sio</code>	Human-gated promotion of reviewer-confirmed candidates into <code>corpus/sios/</code> + <code>corpus/sources/</code> .

3.2 Media, video & timestamp tooling

Tool (npm)	What it does
<code>find-video-sources</code>	Per-SIO media status, missing fields, and search targets for finding official videos.
<code>find-video-moments</code> / <code>find-clip-moments</code>	Surface candidate timestamp moments from chapter markers or fetchable transcripts (report-only).
<code>extract-video-timestamps</code>	ASR caption alignment via <code>yt-dlp</code> ; auto-applies only HIGH-confidence exact-quote matches. The "timestamp embedding agent."
<code>validate-media</code>	Format/consistency/honesty checks on all SIO media metadata; fails hard on violations.
<code>audit-corpus-media</code> / <code>build-media-inventory</code>	Full-corpus media inventories and backfill categorization.
<code>verify-youtube-channels</code>	Resolves official channel IDs into the trusted-channel registry (the only sanctioned source of real channel IDs).
<code>apply-media-promotions</code>	Applies a vetted batch of media-embed promotions via targeted edits.
<code>test-youtube-api</code> / <code>test-media-helpers</code> / <code>test-media-e2e</code>	Tests for the YouTube helper and the media-rendering layer.

3.3 Evaluation, QA & testing

Tool (npm)	What it does
<code>test-retrieval-calibration</code>	Large retrieval + classification regression suite (~100+ cases) with a pass-rate target.
<code>test-sio-retrieval</code> / <code>test-state-classification</code>	Smoke tests for state-filtered retrieval and the classifier.
<code>test-present-insight</code>	Checks presentation output for banned phrases and correct formatting.
<code>test-chat-e2e</code> / <code>test-feedback-loop</code> / <code>test-ui-signal</code>	Agent and feedback integration probes (incl. retry paths).
<code>test-smoke-flows</code> / <code>warm</code>	HTTP smoke tests against a running dev server, plus a cold-start warmer.

3.4 Retrieval & magnet diagnostics

Tool (npm)	What it does
<code>diagnose-magnets</code>	Per-state win-rate, winner margin, and diversity analysis to find "query-magnet" SIOs that dominate unfairly.
<code>test-magnet-risk</code>	Promotion gate: fails if a candidate would win >50% of a state's probes (anti-magnet guardrail).

3.5 Feedback analysis & dev utilities

Tool (npm)	What it does
<code>analyze-feedback</code>	Aggregates feedback events (per-SIO yes rates, retry/dwell stats) and recommends a post-beta build path.
<code>ingest / test-tools / test-agent</code>	Dev helpers: load the vector store, smoke the tools, chat with the agent interactively.

Part 4 • How to Make the Existing Tools Better

Each item is a concrete, evidence-based improvement drawn from the tools' own design docs and known limitations.

4.1 The SIO harvesting / discovery agent

- **The verbatim-verification bottleneck is the #1 scaling limit.** The agent deliberately never sets `transcript_verified: true` — a human must confirm wording. Improvement: integrate official transcript sources (TED, YouTube captions via `yt-dlp`) into the pipeline so the agent can propose a verbatim match with evidence for one-click human confirmation, instead of fully manual checking.
- **TED transcripts are JavaScript-rendered** and aren't retrievable by simple fetch. Improvement: a dedicated TED transcript fetcher (headless render or the TED transcript endpoint) so TED candidates can be auto-verified like other sources.
- **The Source Scout only emits queries; it doesn't search.** Improvement: a supervised web/YouTube search step that actually returns and pre-ranks candidate sources for those queries (still human-gated).
- **The LLM resonance judge isn't calibrated.** Improvement: calibrate the judge against human-rated examples once the corpus passes ~50 SIOs, and track judge-vs-human agreement.
- **Gap detection uses editorial thresholds, not real demand.** Improvement: feed real user queries/feedback into gap detection so we harvest what users actually ask for.

4.2 The timestamp embedding agent

- **Coverage is thin and mostly manual.** ASR auto-apply only fires on HIGH-confidence exact-quote matches; paraphrase SIOs need human listening. Improvement: combine ASR alignment with chapter markers and transcript search into a single confidence-scored moment finder, and expand auto-apply to well-aligned close paraphrases with a clear evidence trail.
- **No closed-loop validation after applying.** Improvement: automatically run `validate-media` and a render check after each timestamp write, and flag any embed that doesn't actually start at the claimed moment.
- **Podcast uploads often lack chapters.** Improvement: prioritize sources with transcripts/chapters during harvesting so the timestamp step has material to work with.

4.3 The retrieval engine & magnet controls

- **Magnets are mainly a "no-hint" phenomenon.** Classifier intake hints already fix most cases; a few benign default-aligned dominators (e.g. Newport, Huberman) remain. Improvement: keep the promotion-time win-rate gate as the primary guard, and add exposure-aware scoring as the corpus grows so no single SIO is over-served.

- **Thresholds were calibrated on a tiny corpus (~19 SIOs).** Improvement: recalibrate similarity thresholds, novelty thresholds, and resonance boosts against the expanded corpus and a real evaluation set.
- **MMR diversity is near-no-op at small sizes.** Improvement: re-evaluate its effect once the corpus is ~111 SIOs.

4.4 The feedback loop

- **"Yes" is ambiguous** (politeness vs. genuine aha); dwell time is the disambiguator. Improvement: keep dwell qualification, and add a light, optional qualitative trigger only if double-rejection stays high after enough sessions.
- **Feedback doesn't yet route into anything automatically.** Improvement: wire `analyze-feedback` output into gap detection and a presentation-review queue (human-reviewed), closing the loop from signal to corpus/presentation change.

4.5 The presentation layer

- **Tier-1 structural wins are scoped but not yet shipped:** a visible source-confidence chip, text-only/no-embed parity, a factual credibility line, and quieting the feedback so the insight is the last beat. Improvement: ship these; they reduce cognitive load and strengthen vicarious credibility at low risk.
- **The card re-parses markdown back into slots,** which is brittle. Improvement: pass derived metadata (confidence, credibility) through the structured channel instead of the markdown string.

Part 5 • Tools We Still Need to Build

Future tool / capability	Why it's needed
Persistent vector store	Today's store is in-memory and rebuilds on boot. A persistent store (pgvector / Pinecone / Chroma) is needed before scale and before any cross-session features.
Automated verbatim / transcript verification	The single biggest manual bottleneck. A transcript-fetch + alignment service would let the discovery agent propose verified excerpts at scale.
Feedback-driven gap detection	Close the loop: use real user queries and rejections to decide what to harvest next, instead of editorial guesses.
Lightweight analytics / metrics dashboard	The beta metrics (landing rate, strong-positive, return rate, double-rejection) need a simple view, not manual log reading.
A/B presentation harness	To resolve the Tier-2 presentation bets (carry line, source-header, gesture) with data rather than debate.
Lightweight personalization	Once there's a user identity and enough signal: remember format/tone/creator preferences to sharpen ranking across sessions.
Hybrid external discovery	A filtered, trust-scored external search layer for moments the curated library doesn't cover — clearly distinguished from curated results.
Safety / boundary classifier	A reliable crisis-signal detector that exits the normal flow and routes to real resources, beyond prompt-level handling.
Ingestion-at-scale pipeline	Discovery → cleaning → auto-tagging → embedding → review as a repeatable service as the library grows past a few hundred SIOs.

Part 6 · Risks, Open Questions & Readiness for User Testing

6.1 Key risks

- **Retrieval quality is the core unvalidated assumption.** Everything depends on whether the system reliably surfaces a genuinely well-matched insight. This must be tested with real users in real stuck moments.
- **Copyright / verbatim integrity.** Legacy SIOs include paraphrase reconstructions; the expansion enforces verified verbatim. The standing posture is metadata + short non-substitutive excerpts + link-out; full text only for licensed/in-house content.
- **Small-corpus evaluation.** Thresholds and diagnostics were tuned on a tiny corpus; conclusions drawn before the expanded corpus is recalibrated may mislead.
- **Trust at first contact.** The user pays a vulnerability cost before seeing value; the first insight has to land and feel human, not algorithmic.

6.2 What must be true before sending to users

- **QA gates (run June 4, 2026):** `validate-media` — PASS, 0 hard violations, 0 warnings. `test-magnet-risk` — exits clean, no blocking magnet (one monitored default-path dominator, Shankar in identity-transition, diversified by intake hints in production). `test-retrieval-calibration` — **25/45 (56%), below the 70% target.** The drop is mostly a *stale-test artifact*: ~20 new SIOs joined the library, and several now win queries whose hand-curated "acceptable answer" sets were written for the pre-expansion corpus (e.g. `brooks-theyre-not-bored`, `shankar-identity-paralysis`, `benshahar-arrival-fallacy`). A minority are genuine new over-winners (notably `brooks-theyre-not-bored` across several engagement-drought probes). Both point to the same fix: refresh the calibration acceptable-sets for the new corpus and retune the resonance/threshold weights. **This recalibration is the one true gate remaining before beta.**
- **Coverage (detect-gaps):** all six states at target (111 total), every statexregister cell populated, only one low-priority advisory (direction-collapse 20 vs momentum-gap 16 — a mild balance note, not a hole).
- The manual smoke test passes end to end (preamble → insight → media → feedback → retry).
- The closed-beta plan and first-three-tester plan are ready (both exist as guides).
- Tier-1 presentation improvements are shipped (recommended, not blocking).

Bottom line. The architecture, tooling, process, and now the corpus (111 verified SIOs, media validated, coverage balanced) are all in place and mature for a v1. **One gate remains before beta:** a retrieval recalibration pass — refresh the calibration acceptable-sets for the expanded corpus and retune the resonance/threshold weights to lift the pass rate back toward the $\geq 70\%$ target and tame the one or two new over-winners. After that, Silhouette is ready for a closed beta whose single job is to validate that the insight reliably lands.

Appendix

A • Glossary

- **SIO (Structured Insight Object)** — the unit of the library: one curated insight + rich metadata.
- **Stuck state** — one of six soft, non-clinical clusters describing the user's moment.
- **Resonance profile** — `insight_type` + `voice_register` ; the second retrieval dimension.
- **Magnet** — an SIO that wins a disproportionate share of a state's queries; guarded by a promotion gate.
- **scoredSearch** — the retrieval function: semantic similarity + metadata boosts + MMR diversity.
- **aha moment** — the single success metric: a real shift in how the user sees their situation.

B • npm command reference (selected)

`discover-sios` , `detect-gaps` , `analyze-user-needs` , `find-source-candidates` , `evaluate-candidate` , `score-candidate-novelty` , `verify-candidate-source` , `review-candidates` , `draft-sio-from-candidate` · `find-video-sources` , `find-video-moments` , `find-clip-moments` , `extract-video-timestamps` , `validate-media` , `audit-corpus-media` , `verify-youtube-channels` · `test-retrieval-calibration` , `test-sio-retrieval` , `test-state-classification` , `test-present-insight` , `test-smoke-flows` · `diagnose-magnets` , `test-magnet-risk` · `analyze-feedback` , `ingest` , `warm` .

C • Key documents

Product: `aiDocs/context.md` , `prd.md` , `mvp.md` , `architecture.md` · System map: `ai/guides/silhouette_system_map.md` · Components: `retrieval ( retrieval_philosophy.md , retrieval_engine.md )` , `presentation ( insight_presentation_layer.md )` , `feedback ( feedback_quality_signal_loop.md )` , `discovery ( agentic_sio_discovery_system_design.md )` , `media ( video_source_finder_design.md , youtube_api_integration_design.md )` · Beta: `closed_beta_plan.md` , `first_three_tester_plan.md` , `manual_beta_smoke_test.md` .